

# Aleena Varghese

AI Engineer | Machine Learning Engineer | Data Scientist

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## Professional Summary

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AI Engineer with hands-on experience building production-grade LLM applications using Retrieval-Augmented Generation (RAG), LangChain, vector databases, embeddings, and prompt engineering. M.Sc. Computer Science (Data Analytics) graduate experienced in Python, FastAPI, Machine Learning, Deep Learning, Computer Vision, and scalable AI application development. Passionate about developing intelligent AI systems through iterative experimentation, evaluation, and production-ready engineering.

## Experience

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### AI Engineer Intern

Thiruvananthapuram, Kerala (Remote)

March 2026 – June 2026

*Lightrains Technolabs*

- Engineered a production-grade multimodal AI assistant using Python, FastAPI, OpenAI APIs, and Retrieval-Augmented Generation (RAG) to enable accurate natural language question answering over technical manuals.
- Designed and implemented an end-to-end RAG pipeline integrating document parsing, knowledge graphs, embeddings, vector search, contextual retrieval, and prompt engineering to improve response relevance and groundedness.
- Built multimodal retrieval workflows using Vision Language Models (VLMs) for semantic image retrieval and visual grounding, enabling intelligent understanding of technical diagrams alongside textual documentation.
- Developed production AI workflows including contextual memory, response validation, AI safety guardrails, confidence scoring, and API integration to improve reliability, explainability, and user experience.

## Projects

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### LLM-powered AI Gmail Assistant | Python, LangChain, Google Gemini, ChromaDB, Sentence Transformers, Streamlit

- Developed an AI-powered Gmail assistant using LangChain, Google Gemini 2.5 Flash, and Retrieval-Augmented Generation (RAG) to answer natural language questions over Gmail.
- Built semantic retrieval using Sentence Transformer embeddings and ChromaDB vector database for context-aware similarity search and persistent memory.
- Implemented email summarization, categorization, and conversational question answering using prompt engineering, embeddings, and retrieval pipelines.
- Integrated Gmail API, LangChain, vector search, and Streamlit into an end-to-end production-ready AI application.

### HR Assistant System | Django, React, TypeScript, Llama 3.3, SQLite

- Developed an AI-powered HR Assistant using Django, React, TypeScript, and Llama 3.3 to automate employee queries related to payroll, leave management, company policies, and HR documentation.
- Designed a contextual AI chatbot using prompt engineering techniques to deliver accurate, role-aware, and policy-based responses.
- Implemented role-based access control, audit logging, secure REST APIs, and modular backend services to improve transparency, security, and maintainability.

### SignBot – Sign Language Recognition | Python, OpenAI CLIP, SVM, Google Gemini AI, Streamlit

- Developed a real-time Sign Language Recognition system using OpenAI CLIP for feature extraction and an SVM classifier for gesture recognition.
- Integrated Google Gemini AI and Text-to-Speech (TTS) to generate context-aware textual and voice responses, improving accessibility for users with hearing and speech impairments.
- Built an end-to-end AI pipeline covering dataset preparation, feature extraction, model training, evaluation, and real-time inference.

**Age, Gender, and Emotion Prediction** | Python, TensorFlow, Keras, OpenCV

- Developed CNN-based deep learning models using TensorFlow/Keras to predict age, gender, and facial emotions from images using the UTKFace and FER2013 datasets.
- Built complete data preprocessing, model training, evaluation, and inference pipelines using TensorFlow, OpenCV, NumPy, Pandas, and Scikit-learn.
- Evaluated model performance using standard machine learning metrics to improve prediction accuracy and model reliability.

**Education**

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|---------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>M.Sc. Computer Science (Data Analytics)</b><br><i>Digital University Kerala</i><br>CGPA: <b>9.12</b> | Thiruvananthapuram, Kerala<br><br>2024 – 2026 |
| <b>B.Sc. Mathematics</b><br><i>Mar Ivanios College (Autonomous)</i><br>CGPA: <b>7.71</b>                | Thiruvananthapuram, Kerala<br><br>2021 – 2024 |

**Technical Skills**

**Programming Languages:** Python, SQL, JavaScript

**Generative AI:** Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain, OpenAI API, Google Gemini API, Embeddings, Semantic Search, ChromaDB, Vector Databases, Knowledge Graphs

**Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), TensorFlow, Keras, Scikit-learn, Support Vector Machines (SVM)

**Frameworks & Libraries:** FastAPI, Django, React, Next.js, Streamlit, OpenCV, NumPy, Pandas, Matplotlib

**Tools:** Git, GitHub, Docker, LightRAG, MinerU, CLIP, Vision Language Models (VLMs), PydanticAI

**Databases:** SQLite, PostgreSQL, ChromaDB, Vector Databases, Knowledge Graph Stores

**Professional Skills**

Problem Solving • Analytical Thinking • Technical Documentation • Adaptability • Continuous Learning

**Certifications**

- Introduction to Large Language Models – Google Cloud
- Google Cloud Career Launchpad – Data Analytics & Computing Foundations
- Python Data Structures – University of Michigan (Coursera)
- Introduction to Git and GitHub – Google (Coursera)
- Data Analytics Job Simulation – Deloitte
- Data Visualization – Tata Group